

Static Testing

Finding faults by reading and reviewing the work — without ever running the software.

Software Inspections · Structured Walkthroughs · Technical Reviews

WHY THIS TOPIC EXISTS

The cheapest bug to fix is the one you catch before the code even runs.

Every other technique we've studied runs the software and watches what happens — that's dynamic testing. But a fault written into a requirements document or a design spec is still a fault, even before a single line of code exists. Static testing catches those faults by examining the work itself: reading, reviewing, and inspecting documents and code, without executing anything.

It is proven to be a cost-effective technique of error detection — fix it on paper, not in production.

Static vs. Dynamic Testing

In static testing, test cases are NOT executed — it does not check the operation or functioning of the software. Instead, each stage of the software life cycle is checked to ensure it follows the specifications, requirements, or standards.

STATIC TESTING

The code is never run.

Action	read, review, inspect
Checks	documents, design, code
Finds	faults before execution
Cost	lowest — caught earliest

DYNAMIC TESTING

The code is executed.

Action	run and observe output
Checks	behaviour at run-time
Finds	faults during execution
Cost	higher — caught later

What Can Be Tested Staticly?

Because nothing is executed, static testing can examine almost any written artefact of a project — long before working software exists.

Requirements

provided by the client

Design specs

of the software

Test plans

before any test runs

Source code

read, not executed

Manuals

user & operational

Procedures & rules

the ways of working

Guidelines

internal best practice

Standard specs

the standards to meet

Three Types of Static Testing

All three are reviews — they differ in how formal they are, who leads them, and how much process they need.

1

Software Inspection

The MOST formal. Led by a moderator, with defined roles and phases.

Most formal

2

Structured Walkthrough

Less formal. Led by the author, who walks peers through the document.

Middle ground

3

Technical Review

Informal. Led by a technical expert, focused on technical content.

Least formal

Software Inspection

Inspection is the process where software documents are checked for deviations from the requirement specification or quality standards. It can be done manually or with tools — but always without executing the software. It is the most formal type of review, carried out by a team of testers.

WHAT GETS INSPECTED

- Software Requirement Specification (SRS)
- Design document
- Source code
- Test plan

KEY IDEA

A structured hunt for defects

Unlike a casual read-through, an inspection has a defined team, defined roles, and a defined sequence of phases. That structure is exactly what makes it the most thorough — and the most reliable at catching defects early.

The Inspection Team — Four Roles

AUTHOR

Creator of the document being inspected.

REVIEWER

Team member responsible for checking the document under test for defects.

MODERATOR

A peer who manages the whole process: schedules the meeting, distributes the document, conducts the discussion, and collects the data.

RECORDER

Records all defects identified, the outcomes of discussions, and any feedback that could improve the document's quality.

The Phases of a Review

An inspection follows a fixed sequence — each phase has a clear job.

1	Planning	Author asks the moderator to review the document; the moderator schedules the meeting and decides pages, team, and approach.
2	Kick-off	All members are brought to the same understanding; the objective and the checking rate (pages per hour) are agreed.
3	Preparation	Each reviewer checks the document alone, noting down every defect, comment, and query.
4	Review Meeting	The team meets — in three parts: Logging (record findings), Discussion (debate them), Decision (is each a valid defect?).
5	Rework	The author reworks the document to resolve all the defects that were confirmed.
6	Follow-up	The moderator follows up to make sure the required corrections were actually done.

Benefits of the Inspection Process

1

Catches bugs at the earliest stage

Defects are found in the documents themselves — preventing all the damage that a defective requirement or test plan would otherwise cause downstream.

2

Reduces cost

A defect identified at the earliest stage is dramatically cheaper to fix than the same defect discovered in finished, running software.

3

Improves future quality

Feedback from the meetings raises the quality of the documents and helps the team avoid similar defects in future work.

Structured Walkthroughs

A walkthrough is less formal than an inspection. It is conducted by the AUTHOR, who walks their peers through the document to gather feedback — giving the author another person's perspective. It can be pre-planned or arranged as needed, and is especially useful for feedback on high-level requirements, and for involving people who are not from the software discipline.

Led by the author · informal · gets a fresh perspective

GOALS OF A WALKTHROUGH

- Gather information by involving stakeholders — inside and outside the software discipline.
- Explain the contents of the document and evaluate it.
- Reach a common consensus about the content.
- Discuss different solutions to a problem, check their validity, and propose alternatives.

Technical Review

A technical review is a static technique conducted to identify errors early in the life cycle, with the focus on the **TECHNICAL CONTENT** of the document under test. It is an informal type of review — led by a technical expert, needing less process and fewer documents. No management participation is required, and more formal characteristics such as checklists or issue logs are optional.

THE THREE TYPES ON A FORMALITY SCALE

Technical Review

Informal · expert-led · technical content

Walkthrough

Semi-formal · author-led · fresh perspective

Inspection

Most formal · moderator-led · defined roles & phases

Same goal — catch defects early by reviewing — at three levels of formality.